

No Unconditional Basis at the Critical First-Difference Besov Endpoint

Candidate partial resolution for arXiv:2002.12917

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Status: candidate substantial partial result, likely valid. For every parameter choice in the source's open range, the space cannot have an unconditional Schauder basis. The proof identifies its Banach envelope with $L_1(I^d)$ and transfers any hypothetical unconditional basis to $L_1(I^d)$. Whether these spaces possess a conditional Schauder basis remains open. Novelty is plausible but not certified.

1 Source question

Let $I = [0, 1]$, let

$$0 < p < q \leq 1, \quad s = d(1/p - 1), \quad (1)$$

and put $X = B_{p,q,1}^s(I^d)$, the Besov space defined using first-order L_p moduli of smoothness. The source asks on PDF p. 2 whether the spaces in (1) possess Schauder bases, or even unconditional Schauder bases, at all [1]:

Theorem 2 *If $0 < p < 1$ in (2) then the Besov space $B_{p,q,1}^s(I^d)$ does not possess non-trivial bounded linear functionals, i.e., $B_{p,q,1}^s(I^d)' = \{0\}$, if and only if $s < d(1/p - 1)$, $0 < q < \infty$ or $s = d(1/p - 1)$, $1 < q < \infty$.*

Consequently, for these parameters $B_{p,q,1}^s(I^d)$ does not have a Schauder basis at all which is a stronger statement than proving that the Haar wavelet system H^d fails to be a Schauder basis. For the parameters

$$0 < p < q \leq 1, \quad s = d(1/p - 1), \quad (3)$$

where according to Theorem 1 H^d is not a Schauder basis in $B_{p,q,1}^s(I^d)$, we have the continuous embedding $B_{p,q,1}^s(I^d) \subset L_1(I^d)$, and thus $L_\infty(I^d) \subset B_{p,q,1}^s(I^d)'$. We do not know if the spaces $B_{p,q,1}^s(I^d)$ satisfying (3) possess (unconditional) Schauder bases at all.

As Schauder basis in Besov spaces $B_{p,q,1}^s(I^d)$, the tensor-product Haar system $\tilde{H}^d$ behaves the same as H^d for $1 \leq p < \infty$ but fails completely for the parameter range $p < 1$. This is the essence of Theorem 5 in Section 5.3

The condition $s < 1/p$ implicit in the source's setup forces $p > (d-1)/d$. The source proves that the Haar system is not a Schauder basis in this range, but also that $X \hookrightarrow L_1(I^d)$ and that dyadic step functions are dense in X . We resolve the parenthesized, unconditional part of the question negatively.

2 Result and idea of the proof

Theorem 1. *For every $d \geq 1$ and every choice of parameters satisfying (1) and $s < 1/p$, the quasi-Banach space $B_{p,q,1}^s(I^d)$ has no unconditional Schauder basis.*

Idea of the Proof

At the critical exponent, the Besov cost of a constant on a dyadic cube is at most its L_1 mass. Splitting a dyadic step function into its individual cells therefore shows that its convexified Besov norm is at most its L_1 norm. The known embedding into L_1 gives the reverse estimate. Consequently the Banach envelope of X is exactly $L_1(I^d)$, up to equivalent norms.

If (x_n) were an unconditional basis of X , all finite coordinate projections would be uniformly bounded. Bounded operators extend to the Banach envelope, so these projections would extend uniformly to L_1 . Their ordinary partial sums converge on the dense subspace X and hence on all of L_1 . Thus (x_n) would be an unconditional Schauder basis of $L_1(I^d)$, contradicting the classical theorem that nonatomic L_1 has no unconditional Schauder basis.

3 The Banach-envelope computation

Let $\mathcal{S}_k$ be the functions constant on the dyadic cubes of side 2^{-k} , and set

$$E_k(f)_p = \inf_{g \in \mathcal{S}_k} \|f - g\|_p.$$

The source's Lemma 1 gives the equivalent quasi-norm

$$\|f\|_X \asymp \left(\|f\|_p^q + \sum_{k=0}^{\infty} (2^{ks} E_k(f)_p)^q \right)^{1/q}. \quad (2)$$

For a quasi-Banach space Z , write

$$\|z\|_c := \inf \left\{ \sum_{j=1}^N \|z_j\|_Z : z = \sum_{j=1}^N z_j, N < \infty \right\}. \quad (3)$$

Its Hausdorff completion is the Banach envelope whenever (3) separates points.

Lemma 2 (Critical dyadic atoms). *If Δ is a dyadic cube of level m and a is a scalar, then*

$$\|a\mathbf{1}_\Delta\|_X \lesssim |a| |\Delta|. \quad (4)$$

The implicit constant is independent of a, m , and Δ .

Proof. For $k \geq m$, the atom belongs to $\mathcal{S}_k$, so its best-approximation error is zero. For $k < m$,

$$E_k(a\mathbf{1}_\Delta)_p \leq \|a\mathbf{1}_\Delta\|_p = |a| |\Delta|^{1/p}.$$

Using (2), $s > 0$, and the geometric sum,

$$\begin{aligned} \|a\mathbf{1}_\Delta\|_X &\lesssim |a| |\Delta|^{1/p} \left(1 + \sum_{k < m} 2^{ksq} \right)^{1/q} \\ &\lesssim |a| |\Delta|^{1/p} 2^{ms}. \end{aligned}$$

Since $|\Delta| = 2^{-md}$ and $s = d(1/p - 1)$, the last product is $|a| 2^{-md/p} 2^{md(1/p-1)} = |a| 2^{-md} = |a| |\Delta|$. The unweighted L_p term is smaller because $|\Delta|^{1/p} \leq |\Delta|$ for $p < 1$. $\square$

Lemma 3 (Envelope identification). *The canonical inclusion of X into $L_1(I^d)$ extends to an isomorphism from the Banach envelope $\widehat{X}$ onto $L_1(I^d)$.*

Proof. The source proves $\|f\|_1 \lesssim \|f\|_X$ in the range (1). Hence every finite decomposition in (3) gives

$$\|f\|_1 \leq \sum_j \|f_j\|_1 \lesssim \sum_j \|f_j\|_X,$$

and therefore $\|f\|_1 \lesssim \|f\|_c$. In particular, (3) separates points.

If $f = \sum_{\Delta \in \mathcal{T}_m} a_\Delta \mathbf{1}_\Delta$ is a dyadic step function on one level, Lemma 2 and this very decomposition give

$$\|f\|_c \leq \sum_{\Delta \in \mathcal{T}_m} \|a_\Delta \mathbf{1}_\Delta\|_X \lesssim \sum_{\Delta} |a_\Delta| |\Delta| = \|f\|_1.$$

Thus the envelope norm and L_1 norm are equivalent on all dyadic step functions. They are dense in X by the source and dense in $L_1(I^d)$. Completing proves the claim. $\square$

4 Unconditional bases pass to the envelope

We record the required transfer without invoking a black-box envelope theorem.

Lemma 4. *Let a quasi-Banach space Z inject continuously and densely into its Banach envelope $\widehat{Z}$. If (z_n) is an unconditional Schauder basis of Z , then (z_n) is an unconditional Schauder basis of $\widehat{Z}$.*

Proof. Every bounded operator $T : Z \rightarrow Z$ is bounded for the convexified norm: for every finite decomposition $z = \sum_j z_j$,

$$\|Tz\|_c \leq \sum_j \|Tz_j\|_Z \leq \|T\| \sum_j \|z_j\|_Z.$$

It consequently extends to $\widehat{Z}$.

Let P_A be the coordinate projection onto a finite set $A \subset \mathbb{N}$. Unconditionality gives $\sup_A \|P_A\| < \infty$, so all P_A extend to $\widehat{Z}$ with a common bound. In particular, the ordinary partial sums $P_{[1,N]}z$ converge to z for $z \in Z$. Uniform boundedness and density then imply convergence for every $z \in \widehat{Z}$: approximate z by $y \in Z$ and estimate

$$\|P_{[1,N]}z - z\| \leq \|P_{[1,N]}(z - y)\| + \|P_{[1,N]}y - y\| + \|y - z\|.$$

Each extended $P_{[1,N]}$ still has range $\text{span}\{z_1, \dots, z_N\}$, since that finite-dimensional range is closed. The projection identities persist by continuity, so these partial sums give a Schauder basis of $\widehat{Z}$. The uniform bounds for all finite P_A give unconditionality there. $\square$

Proof of Theorem 1. By Lemma 3, $\widehat{X} \simeq L_1(I^d)$. An unconditional basis of X would, by Lemma 4, induce an unconditional basis of $L_1(I^d)$. This contradicts the classical nonexistence theorem for unconditional Schauder bases in nonatomic L_1 ; the source itself recalls this theorem and cites Kashin–Saakyan for it. Therefore X has no unconditional Schauder basis. $\square$

5 Why the conditional question remains

For an ordinary Schauder basis, only the initial coordinate projections are uniformly bounded. The proof of Lemma 4 still shows that such a basis of X would pass to a conditional basis of L_1 . This is no contradiction: $L_1(I^d)$ does have conditional Schauder bases, including a suitable ordering of the Haar system. Thus the envelope method separates the two words in the source’s “(unconditional)” question sharply.

Standard smooth-wavelet results do not fill the gap: at the endpoint (1), the first-difference space $B_{p,q,1}^s$ is not covered by the strict norm-equivalence range with the distributional Besov space. Likewise, higher-order spline results concern spaces defined by higher-order moduli. Finally, the source’s exponential lower bound for the norms of the dyadic conditional expectations rules out the natural Haar filtration, but it does not rule out unrelated finite-rank approximation schemes. A resolution of the conditional-basis question therefore requires a new construction or a genuinely stronger invariant than the Banach envelope.

References

- [1] P. Oswald, *Multivariate Haar systems in Besov function spaces*, Sbornik Math. **212** (2021), 810–842; [arXiv:2002.12917](#).